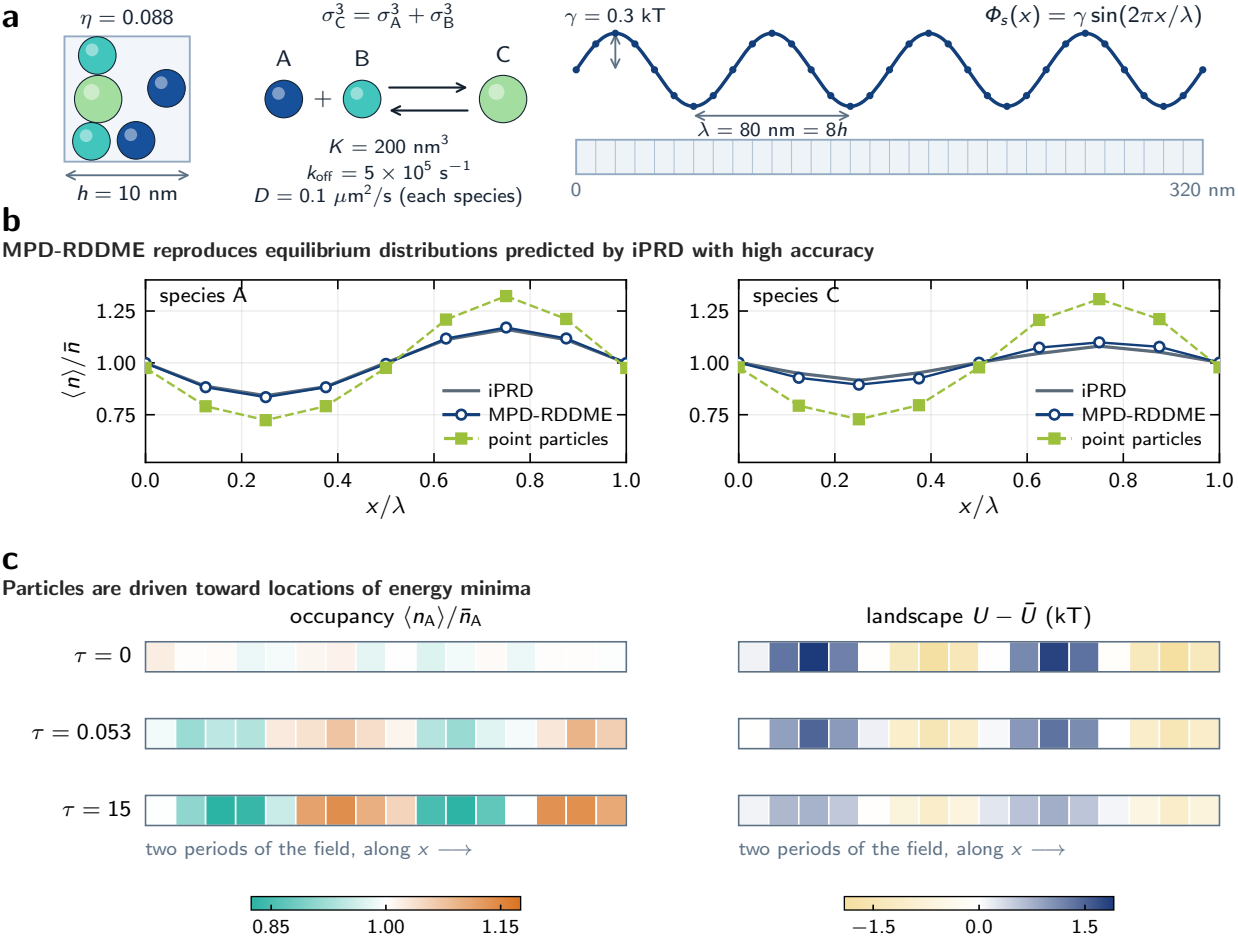

a
$\eta = 0.088$
$h = 10$ nm
$\sigma_C^3 = \sigma_A^3 + \sigma_B^3$
A
B
C
$K = 200$ nm^3
$k_{off} = 5 \times 10^5$ s^{-1}
$D = 0.1$ μm^2/s (each species)
$\gamma = 0.3$ kT
$\Phi_s(x) = \gamma \sin(2\pi x/\lambda)$
$\lambda = 80$ nm $= 8h$
0
320 nm
b
MPD-RDDME reproduces equilibrium distributions predicted by iPRD with high accuracy
species A
species C
$\langle n \rangle / \bar{n}$
1.25
1.00
0.75
0.0
0.2
0.4
0.6
0.8
1.0
x/λ
iPRD
MPD-RDDME
point particles
c
Particles are driven toward locations of energy minima
occupancy $\langle n_A \rangle / \bar{n}_A$
landscape $U - \bar{U}$ (kT)
$\tau = 0$
$\tau = 0.053$
$\tau = 15$
two periods of the field, along x ⟶
two periods of the field, along x ⟶
0.85
1.00
1.15
−1.5
0.0
1.5